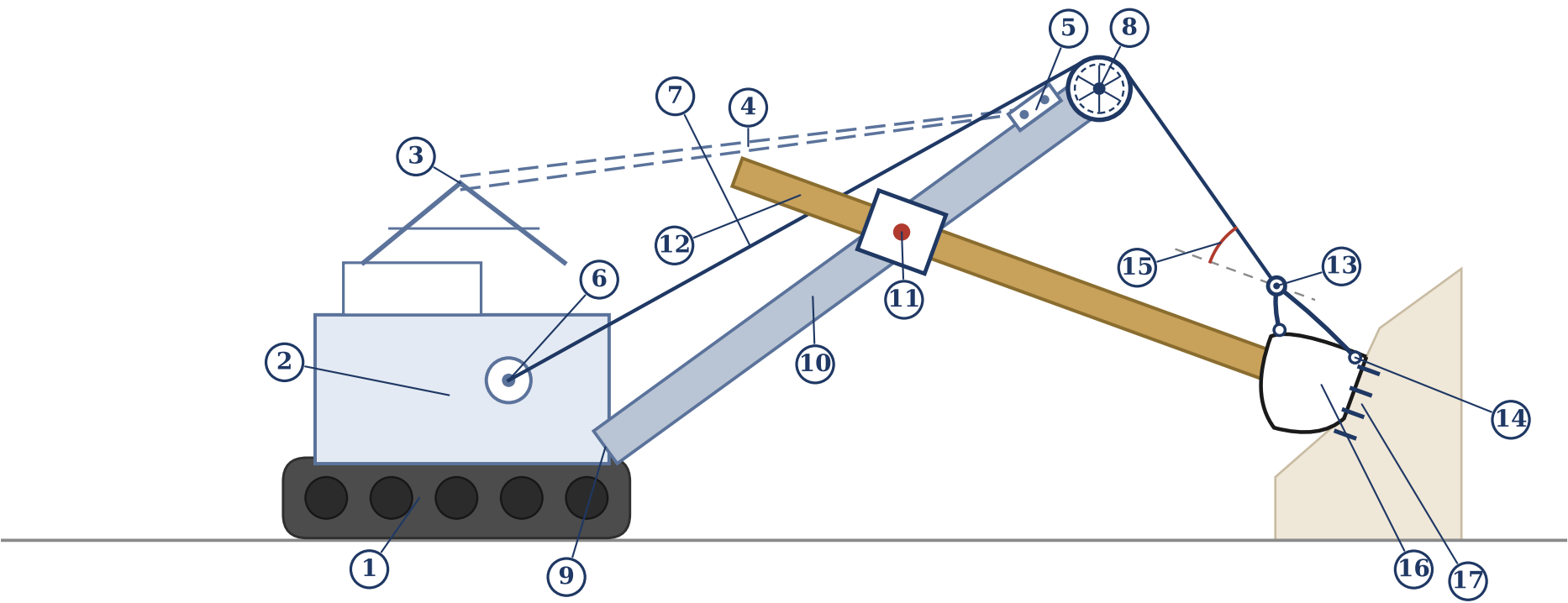


(a) Machine anatomy

the Stage-1 model covers the saddle pivot, the dipper handle and the dipper (coloured); boom, gantry, ropes and swing are shown for context and enter at later milestones



Key

- 1 crawler tracks (lower works)
- 2 house / upper works (swing frame)
- 3 gantry (A-frame)
- 4 suspension (pendant) ropes
- 5 boom-head connector
- 6 hoist drum
- 7 hoist rope
- 8 boom-point sheave
- 9 boom foot (pinned to the house)
- 10 boom
- 11 saddle block (revolute pivot, q_3)
- 12 dipper handle (prismatic crowd, d_4)
- 13 dipper bail (yoke and rope socket)
- 14 bail pins in the dipper lugs
- 15 α_3 (rope angle at the bail)
- 16 dipper
- 17 dipper teeth (bank interface, where the digging force acts)

Coloured members are modelled at Stage 1; greyed members enter at later milestones.